

Chemical Rockets

Single stage rocket

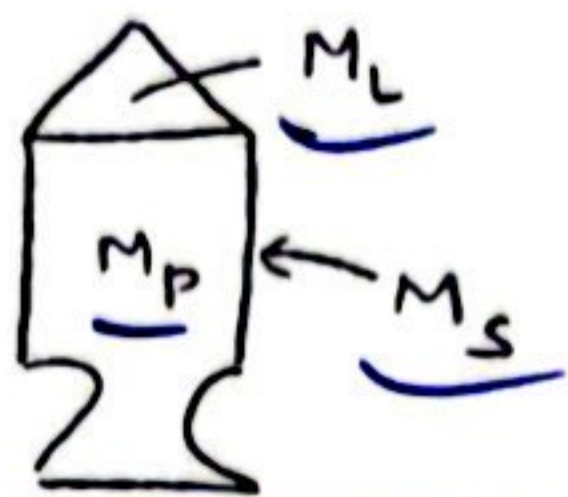
M_L - payload mass

M_S - structural mass

M_P - propellant mass

M_0 - initial mass

M_b - burnout mass



$$M_0 = M_L + M_S + M_P \quad (10.19)$$

$$M_b = M_0 - M_P = M_L + M_S \quad (10.20)$$

$$\lambda \stackrel{\text{def}}{=} \frac{M_L}{M_0 - M_L} \quad (10.22) = \frac{M_L}{M_S + M_P} \quad \lambda \ll 1$$

payload ratio

$$\epsilon \stackrel{\text{def}}{=} \frac{M_S}{M_S + M_P} = \frac{M_S}{M_0 - M_L} \quad (10.23) \quad \epsilon < 1$$

structural coefficient

$$R = \frac{M_0}{M_b} = \frac{1 + \lambda}{\epsilon + \lambda} \quad (10.24)$$

mass ratio

verify

$$\Delta u_{b, \text{imp}} = u_{eq} \ln \left(\frac{M_0}{M_b} \right) = u_{eq} \ln \frac{1 + \lambda}{\epsilon + \lambda} = I_{sp} g_0 \ln \frac{1 + \lambda}{\epsilon + \lambda}$$

$$\left\{ \text{Recall } \Delta u_b = \Delta u_{b, \text{imp}} - \overline{g \cos \theta} t_b - \left(\frac{\overline{g}}{\overline{M}} \right) t_b \right\}$$

Multistage rocket

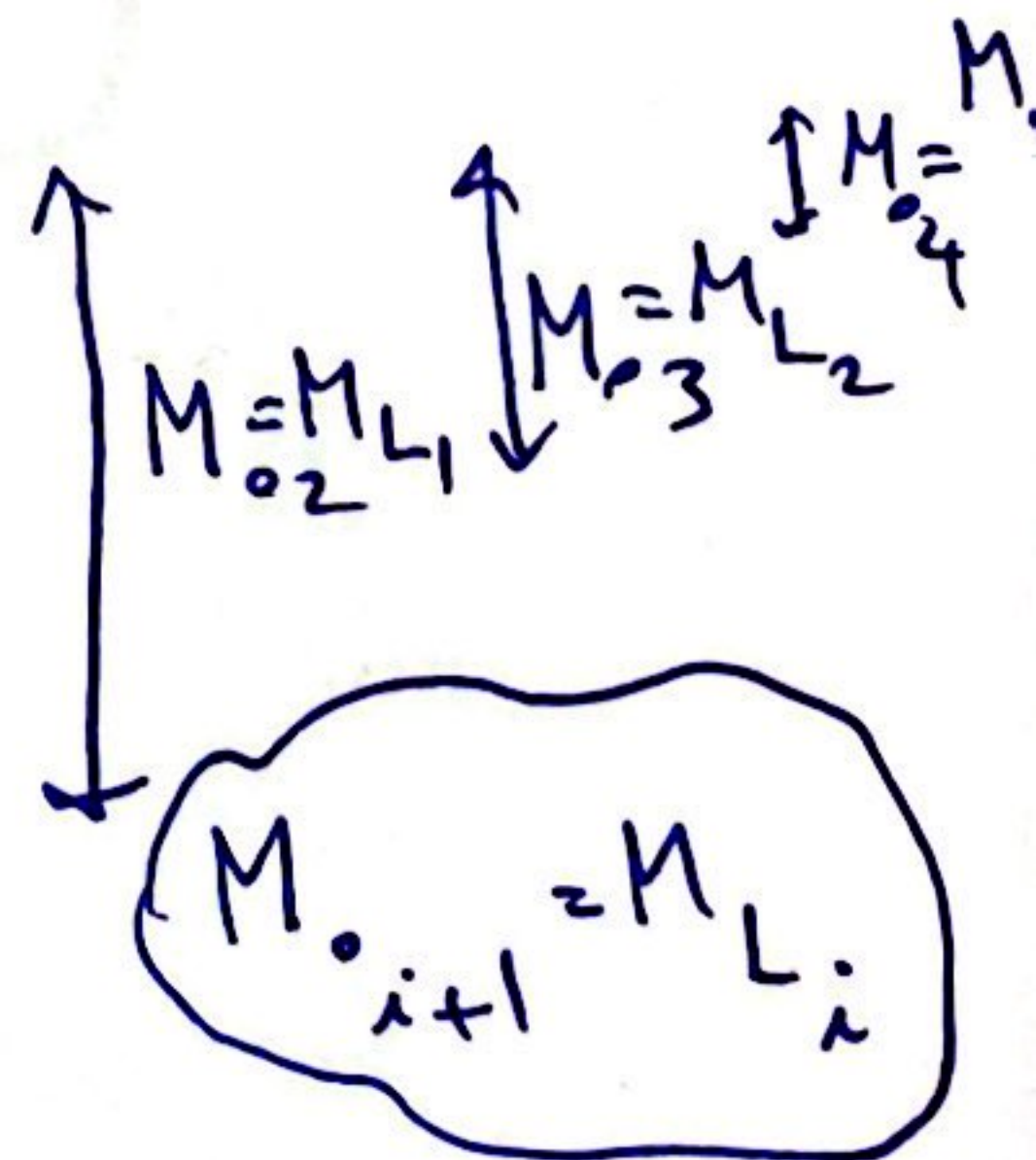
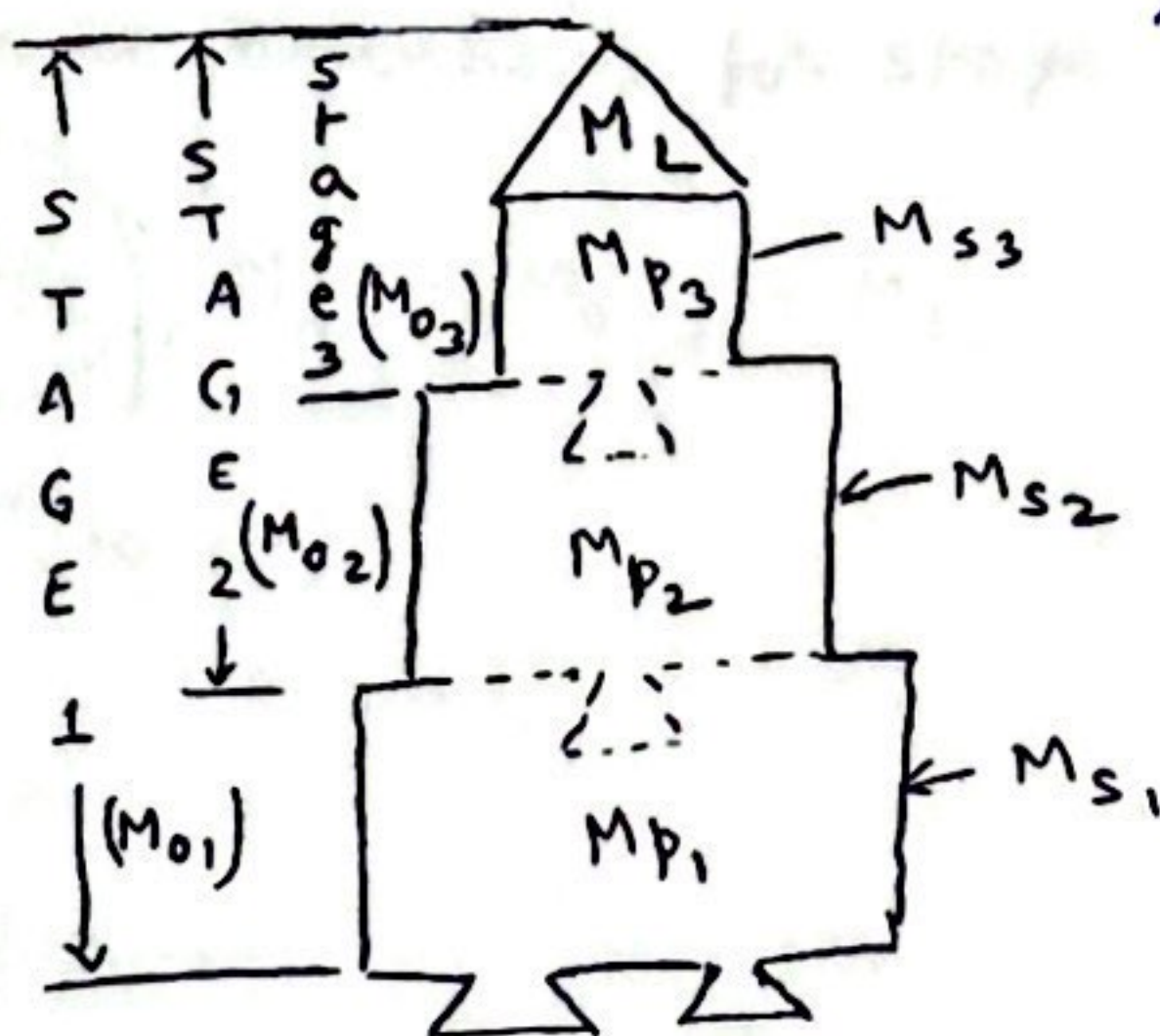
stage 1:

$$M_{01} = M_{L1} + M_{S1} + M_{P1}$$

The payload mass for stage 1 is the initial mass of stage 2.

$$M_{01} = M_{02} + M_{S1} + M_{P1}$$

$$M_{0i} = M_{0i+1} + M_{S_i} + M_{P_i}$$



$$M_{b1} = M_{o1} - M_{p1} = M_{o2} + M_{s1}$$

$$Q_1 = \frac{M_{o1}}{M_{b1}}$$

$$\lambda_1 = \frac{M_{L1}}{M_{o1} - M_{L1}} = \frac{M_{o2}}{M_{o1} - M_{o2}}, \quad \epsilon_1 = \frac{M_{s1}}{M_{s1} + M_{p1}} = \frac{M_{s1}}{M_{o1} - M_{L1}} = \frac{M_{s1}}{M_{o1} - M_{o2}}$$

$$\Delta u_{b,imp1} = u_{eq1} \ln Q_1 = u_{eq1} \ln \frac{1 + \lambda_1}{\epsilon_1 + \lambda_1}$$

stage 1

Similarly, for stage 2,

$$M_{o2} = M_{o3} + M_{s2} + M_{p2}, \quad M_{b2} = M_{o3} + M_{s2}$$

$$Q_2 = \frac{M_{o2}}{M_{b2}}, \quad \lambda_2 = \frac{M_{o3}}{M_{o2} - M_{o3}}, \quad \epsilon_2 = \frac{M_{s2}}{M_{o2} - M_{o3}}$$

$$\Delta u_{b,imp2} = u_{eq2} \ln Q_2 = u_{eq2} \ln \frac{1 + \lambda_2}{\epsilon_2 + \lambda_2}$$

stage 2

and, for stage 3,

$$M_{o3} = M_{o4} + M_{s3} + M_{p3}, \quad M_{b3} = M_{o4} + M_{s3}$$

$$Q_3 = \frac{M_{o3}}{M_{b3}}, \quad \lambda_3 = \frac{M_{o4}}{M_{o3} - M_{o4}}, \quad \epsilon_3 = \frac{M_{s3}}{M_{o3} - M_{o4}}$$

$$\Delta u_{b,imp3} = u_{eq3} \ln Q_3 = u_{eq3} \ln \frac{1 + \lambda_3}{\epsilon_3 + \lambda_3}$$

$$\Delta u_{b,imp tot} = u_{eq1} \ln \frac{1 + \lambda_1}{\epsilon_1 + \lambda_1} + u_{eq2} \ln \frac{1 + \lambda_2}{\epsilon_2 + \lambda_2} + u_{eq3} \ln \frac{1 + \lambda_3}{\epsilon_3 + \lambda_3}$$

The text provides the same equations for stage i

$$M_{oi} = M_{oi+1} + M_{si} + M_{pi}, \quad M_{bi} = M_{oi+1} + M_{si}$$

$$\lambda_i = \frac{M_{oi+1}}{M_{oi} - M_{oi+1}} \quad (10.27)$$

$$\epsilon_i = \frac{M_{si}}{M_{oi} - M_{oi+1}} \quad (10.28)$$

$$\sum_1^3 Q = Q + Q + Q$$

$$R_i = \frac{1 + \lambda_i}{\epsilon_i + \lambda_i}$$

$$\Delta u_{b, \text{imp tot}} = \sum_{i=1}^n u_{eq_i} \ln \left(\frac{1 + \lambda_i}{\epsilon_i + \lambda_i} \right) = \sum_{i=1}^n \overbrace{I_{sp_i} g_e}^{u_{eq_i}} \ln \left(\frac{1 + \lambda_i}{\epsilon_i + \lambda_i} \right) \quad (10.31) \quad \# \text{ of stages}$$

ASIDE (see page 484, text)

$$\lambda_1 = \frac{M_{02} \approx M_L}{M_{01} - M_{02}}$$

$$1 + \lambda_1 = 1 + \frac{M_{02} = M_L}{M_{01} - M_{02}} = \frac{M_{01}}{M_{01} - M_{02}}$$

$$\Rightarrow \frac{1 + \lambda_1}{\lambda_1} = \frac{M_{01}}{M_{02}}$$

Similarly,

$$\frac{1 + \lambda_2}{\lambda_2} = \frac{M_{02}}{M_{03}}$$

$$\frac{1 + \lambda_3}{\lambda_3} = \frac{M_{03}}{\cancel{M_{04}} \rightarrow M_L}$$

Combining,

$$\left(\frac{1 + \lambda_1}{\lambda_1} \right) \left(\frac{1 + \lambda_2}{\lambda_2} \right) \left(\frac{1 + \lambda_3}{\lambda_3} \right) = \frac{M_{01}}{\cancel{M_{02}}} \cdot \frac{\cancel{M_{02}}}{\cancel{M_{03}}} \cdot \frac{\cancel{M_{03}}}{M_L} = \frac{M_{01}}{M_L}$$

The above equation is for a three-stage rocket. In general,

$$\frac{M_{01}}{M_L} = \prod_{i=1}^n \left(\frac{1 + \lambda_i}{\lambda_i} \right) \quad (10.35)$$

↑

Ex: Consider a three-stage rocket with the following

characteristics:

	Stage 1	Stage 2	Stage 3
M_0 (kg)	2,500,000	580,000	179,000
M_p (kg)	1,824,000	373,000	90,000
M_s (kg)	96,000	28,000	21,000
I_{sp} (s)	370	370	370
u_{eq} ($\frac{m}{s}$)	3629.7	3629.7	3629.7
M_L (kg)	580,000	179,000	68000
λ	0.3021	0.4464	0.6126
ϵ	0.05	0.0698	0.1892
$\Delta u_{b,imp}$ ($\frac{m}{s}$)	4747	3740	2536

$$\Delta u_{b,imp tot} = (\Delta u_{b,imp1} + \Delta u_{b,imp2} + \Delta u_{b,imp3})$$

$$= 11023 \text{ m/s}$$

{ In the table, given values are in black,
determined values are in red }

Now, consider a single stage rocket with the same
payload, structural and propellant masses as the
three-stage rocket, above.

$$M_0 = 2,500,000 \text{ kg}$$

$$M_s = (96,000 + 28,000 + 21,000) = 145,000 \text{ kg}$$

$$M_p = (1,824,000 + 373,000 + 90,000) = 2,287,000 \text{ kg}$$

$$u_{eq} = I_{sp} g_e$$

$$\lambda_i = \frac{M_{Li}}{M_{0i} - M_{Li}}$$

$$\epsilon_i = \frac{M_{si}}{M_{si} + M_{pi}}$$

$$\Delta u_{b,imp i} = u_{eq i} \ln Q_i$$

$$M_L = 68,000 \text{ kg}$$

$$I_{sp} = 370 \text{ s}, u_{eq} = 3629.7 \text{ m/s}$$

$$\lambda = \frac{M_L}{M_0 - M_L} = 0.02796, \epsilon = \frac{M_s}{M_0 - M_L} = 0.05962$$

$$\Delta u_{b,imp} = u_{eq} \ln \frac{1+\lambda}{\epsilon+\lambda} = 8939 \text{ m/s}$$

Staging has increased the velocity rise by 23.3%.

$$\left\{ \frac{11023 - 8939}{8939} \otimes 100 \right\}$$

Optimization of multistage rocket

Known: $\Delta u_{b,imp}$ (from mission calculation)

Assumption: The number of stages, and the

I_{sp} and ϵ values of each are known.

Problem: To find the values of λ_i such that

The $\left(\frac{M_L}{M_{01}}\right)$ ratio is maximized.

$$\Delta u_{b,imp} = \sum_{i=1}^n g_e I_{sp_i} \ln \left(\frac{1+\lambda_i}{\epsilon_i+\lambda_i} \right) \begin{cases} I_{sp_i} \text{ and } \epsilon_i \\ \text{are known, see} \\ \text{assumption} \end{cases}$$

The value of $\Delta u_{b,imp}$ is known. Thus, the various λ_i should be changed in such a way that $\Delta u_{b,imp}$ stays fixed in the maximization of $\left(\frac{M_L}{M_{01}}\right)$. This is known as a constrained optimization problem. The constraint is mathematically written as

$$\delta [\Delta u_{b,imp}] = 0$$

$$\Rightarrow \delta \left[\sum_{i=1}^n g_e I_{sp_i} \ln \left(\frac{1+\lambda_i}{\epsilon_i+\lambda_i} \right) \right] = 0$$

$$\Rightarrow \delta \left[\sum_{i=1}^n g_e I_{sp_i} \ln \left(\frac{1+\lambda_i}{\epsilon_i+\lambda_i} \right) \right] = 0$$

$$(or) \sum_{i=1}^n \delta \left[g_e I_{sp_i} \ln \left(\frac{1+\lambda_i}{\epsilon_i+\lambda_i} \right) \right] = 0$$

$$(or) \sum_{i=1}^n \frac{\partial}{\partial \lambda_i} \left[g_e I_{sp_i} \ln \left(\frac{1+\lambda_i}{\epsilon_i+\lambda_i} \right) \right] \delta \lambda_i = 0$$

$$\Rightarrow \sum_{i=1}^n g_e I_{sp_i} \frac{(\epsilon_i - 1)}{(1+\lambda_i)(\epsilon_i+\lambda_i)} \delta \lambda_i = 0 \quad (A)$$

Consider, now, the expression for $\left(\frac{M_{01}}{M_L} \right)$ [Eqn 10.35],

which was obtained earlier. This can be written as

$$\Rightarrow \frac{M_L}{M_{01}} = \prod_{i=1}^n \left(\frac{\lambda_i}{1+\lambda_i} \right)$$

The λ_i are chosen to maximize $\left(\frac{M_L}{M_{01}} \right)$:

$$\delta \left(\frac{M_L}{M_{01}} \right) = 0 \Rightarrow \delta \left[\prod_{i=1}^n \left(\frac{\lambda_i}{1+\lambda_i} \right) \right] = 0$$

$$or, \text{ equivalently, } \delta \left[\sum_{i=1}^n \ln \left(\frac{\lambda_i}{1+\lambda_i} \right) \right] = 0$$

$$or, \sum_{i=1}^n \delta \left[\ln \left(\frac{\lambda_i}{1+\lambda_i} \right) \right] = 0$$

$$or, \sum_{i=1}^n \frac{\partial}{\partial \lambda_i} \left[\ln \left(\frac{\lambda_i}{1+\lambda_i} \right) \right] = 0$$

$$or, \sum_{i=1}^n \frac{1}{\lambda_i (1+\lambda_i)} \delta \lambda_i = 0 \quad (B)$$

Introduce Lagrange multiplier α : undetermined

Lagrange Multiplier :- Maximum of f with $g=0$.

Let $z = f(x, y)$ for maxima or minima

$$dz = 0$$

$$\frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = 0$$

where dx & dy are independent & not zero, it follows that

extremum exist, $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} = 0$

now let $g(x, y) = 0$ be a curve in xy plane. let us find points on the curve $g=0$ at which f is stationary

Constraints $\rightarrow \because g=0 \Rightarrow dg=0$

$$\frac{\partial g}{\partial x} dx + \frac{\partial g}{\partial y} dy = 0$$

$$\frac{dy}{dx} = -\frac{\partial f / \partial x}{\partial f / \partial y} = -\frac{\partial g / \partial x}{\partial g / \partial y}$$

$$\frac{\partial f / \partial x}{\partial g / \partial x} = \frac{\partial f / \partial y}{\partial g / \partial y} = -\alpha$$

Lagrange function:
 $\alpha g + f = h$

$$\alpha g(x, y) + f(x, y) = h(x, y, \alpha) \quad \frac{\partial f}{\partial x} + \alpha \frac{\partial g}{\partial x} = 0 ; \quad \frac{\partial f}{\partial y} + \alpha \frac{\partial g}{\partial y} = 0$$

But these, together with constraint $g(x, y) = 0$ are the condition required for the function

$$h(x, y, \alpha) = f(x, y) + \alpha g(x, y)$$

$$\frac{\partial h}{\partial x} = \frac{\partial f}{\partial x} + \alpha \frac{\partial g}{\partial x} = 0$$

$$\frac{\partial h}{\partial y} = \frac{\partial f}{\partial y} + \alpha \frac{\partial g}{\partial y} = 0$$

$$\frac{\partial h}{\partial \alpha} = g = 0$$

Optimization of multistage Rocket:-

$$\Delta u_{b,imp,tot} = \sum_{i=1}^n g \underbrace{I_{sp,i}}_{\text{Cons}} \underbrace{\ln\left(\frac{1+\lambda_i}{\epsilon_i+\lambda_i}\right)}_{\text{fixed}}$$

for maximization of $\frac{M_L}{M_{01}}$

$$\delta \left[\sum_{i=1}^n g e^{I_{sp,i}} \ln\left(\frac{1+\lambda_i}{\epsilon_i+\lambda_i}\right) \right] = 0$$

fixed

Maximize λ_i

$$\delta \left[\Delta u_{b,imp} \right] = 0$$

$$\delta \left[\sum_{i=1}^n g e^{I_{sp,i}} \ln\left(\frac{1+\lambda_i}{\epsilon_i+\lambda_i}\right) \right] = 0$$

$$\Rightarrow \sum_{i=1}^n \delta \left[g e^{I_{sp,i}} \ln\left(\frac{1+\lambda_i}{\epsilon_i+\lambda_i}\right) \right] = 0$$

$$\delta = \left(\frac{d}{d\lambda_i} \right) d\lambda_i$$

$$\Rightarrow \sum_{i=1}^n \frac{d}{d\lambda_i} \left[g e^{I_{sp,i}} \ln\left(\frac{1+\lambda_i}{\epsilon_i+\lambda_i}\right) \right] d\lambda_i = 0$$

$$\sum_{i=1}^n g e^{I_{sp,i}} \frac{\epsilon_i + \lambda_i}{1 + \lambda_i} \left[\frac{\epsilon_i + \lambda_i - (1 + \lambda_i)}{(\epsilon_i + \lambda_i)^2} \right] d\lambda_i = 0$$

$$\therefore \sum_{i=1}^n g e^{I_{sp,i}} \frac{\epsilon_i - 1}{(1 + \lambda_i)(\epsilon_i + \lambda_i)} d\lambda_i = 0$$

→ (A)

Constraint (A)

fixed

$$\frac{M_L}{M_{01}} = \prod_{i=1}^n \left(\frac{\lambda_i}{1 + \lambda_i} \right)$$

$$\delta \left(\frac{M_L}{M_{01}} \right) = 0$$

$$\Rightarrow \delta \left[\prod_{i=1}^n \left(\frac{\lambda_i}{1 + \lambda_i} \right) \right] = 0$$

(or)

$$\sum_{i=1}^n \delta \left[\ln \left(\frac{\lambda_i}{1 + \lambda_i} \right) \right] =$$

$$\sum_{i=1}^n \frac{1}{\lambda_i (1 + \lambda_i)} d\lambda_i = 0$$

→ (B)

Constraint (B)

if we use Lagrange multiplier for (A) + (B) then

(B) + α (A) = 0 to get ~~minima~~ ^{Max} λ_i

$$\left(\sum_{i=1}^n \frac{1}{\lambda_i (1+\lambda_i)} \delta \lambda_i \right) + \alpha \left(\sum_{i=1}^n g_e I_{sp_i} \frac{\epsilon_i - 1}{(1+\lambda_i)(\epsilon_i + \lambda_i)} \delta \lambda_i \right) = 0$$

Here λ_i is independent

$$\frac{1}{\lambda_i (1+\lambda_i)} + \alpha g_e I_{sp_i} \frac{\epsilon_i - 1}{(1+\lambda_i)(\epsilon_i + \lambda_i)} = 0 \rightarrow (C)$$

$$\frac{1}{\lambda_i (1+\lambda_i)} = \frac{(1-\epsilon_i) \alpha g_e I_{sp_i}}{(1+\lambda_i)(\epsilon_i + \lambda_i)}$$

$$\frac{\epsilon_i + \lambda_i}{\lambda_i} = (1-\epsilon_i) \alpha I_{sp_i} g_e$$

$$\frac{\epsilon_i}{\lambda_i} + 1 = \alpha I_{sp_i} g_e (1-\epsilon_i)$$

$$\frac{\epsilon_i}{\lambda_i} = \alpha I_{sp_i} g_e (1-\epsilon_i) - 1$$

$$\text{Max } \lambda_i \leftarrow \lambda_i = \frac{\epsilon_i}{\alpha g_e I_{sp_i} (1-\epsilon_i) - 1} \rightarrow (D)$$

but there is unknown α

' α ' is obtained from the eq $\Delta U_{b,imp}$

$$\Delta U_{b,imp} = \sum_{i=1}^n g_e I_{sp_i} \ln \left(\frac{1+\lambda_i}{\epsilon_i + \lambda_i} \right)$$